

```
Go Run Terminal Help c assingement 3
C q1.c M X
assignment-3-sshahi7417-blip > src > C q1.c > main()
1 // Write a C program to find the maximum and minimum elements in an array.
2
3
4 #include <stdio.h>
5
6 int main(){
7
8     int arr[] = {12, 45, 2, 67, 34, 89, 5, 23};
9     int n = sizeof(arr) / sizeof(arr[0]);
10
11     int max = arr[0];
12     int min = arr[0];
13
14     for (int i = 1; i < n; i++){
15         if (arr[i] > max){
16             max = arr[i];
17         } else if (arr[i] < min){
18             min = arr[i];
19         }
20     }
21
22     printf("Maximum element: %d\n", max);
23     printf("Minimum element: %d\n", min);
24
25     return 0;
26 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> .\q1.exe
Maximum element: 89
Minimum element: 2
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```
C q1.c M C q2.c M X
assignment-3-sshahi7417-blip > src > C q2.c > ...
1 // Implement a C program to reverse the elements of an array.
2 #include <stdio.h>
3
4 int main() {
5     int arr[] = {10, 20, 30, 40, 50};
6     int n = sizeof(arr) / sizeof(arr[0]);
7     int start = 0;
8     int end = n - 1;
9     int temp;
10
11     while (start < end) {
12         temp = arr[start];
13         arr[start] = arr[end];
14         arr[end] = temp;
15
16         start++;
17         end--;
18     }
19
20     printf("Reversed array: ");
21     for (int i = 0; i < n; i++) {
22         printf("%d ", arr[i]);
23     }
24     printf("\n");
25
26     return 0;
27 }
```

Output

```
> gcc q2.c -o q2
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q2
Reversed array: 50 40 30 20 10
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> 
```

```
C q1.c M C q2.c M C q3.c M X
assignment-3-sshahi7417-blip > src > C q3.c > main()
1 // Write a C program to sort an array of integers in ascending order using a sorting algorithm of your choice (e.g., bubble sort, selection sort)
2
3 #include <stdio.h>
4
5 int main() {
6     int arr[] = {64, 34, 25, 12, 22, 11, 90};
7     int n = sizeof(arr) / sizeof(arr[0]);
8     int temp;
9
10    for (int i = 0; i < n - 1; i++) {
11        for (int j = 0; j < n - i - 1; j++) {
12            if (arr[j] > arr[j + 1]) {
13                temp = arr[j];
14                arr[j] = arr[j + 1];
15                arr[j + 1] = temp;
16            }
17        }
18    }
19
20    printf("Sorted array: ");
21    for (int i = 0; i < n; i++) {
22        printf("%d ", arr[i]);
23    }
24
25    printf("\n");
26
27    return 0;
28 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q3.c -o q3
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q3
Sorted array: 11 12 22 25 34 64 90
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```
C q1.c M C q2.c M C q3.c M C q4.c M X
assignment-3-sshahi7417-blip > src > C q4.c > main()
1 // Implement a C program to find the second largest element in an array.
2
3 #include <stdio.h>
4 #include <limits.h>
5
6 int main() {
7     int arr[] = {12, 35, 1, 10, 34, 1};
8     int n = sizeof(arr) / sizeof(arr[0]);
9
10    if (n < 2) {
11        printf("Array size must be at least 2.\n");
12        return 1;
13    }
14
15    int first = INT_MIN;
16    int second = INT_MIN;
17
18    for (int i = 0; i < n; i++) {
19        if (arr[i] > first) {
20            second = first;
21            first = arr[i];
22        }
23        else if (arr[i] > second && arr[i] != first) {
24            second = arr[i];
25        }
26    }
27
28    if (second == INT_MIN) {
29        printf("No second largest element found.\n");
30    } else {
31        printf("The second largest element is %d\n", second);
32    }
33
34    return 0;
35 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q4.c -o q4
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q4
The second largest element is 34
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```
assignment-3-ssahi7417-blip > src > C q5.c > main()
1 // Write a C program to merge two sorted arrays into a single sorted array.
2
3 #include <stdio.h>
4
5 void mergeArrays(int arr1[], int n1, int arr2[], int n2, int merged[]) {
6     int i = 0, j = 0, k = 0;
7
8     while (i < n1 && j < n2) {
9         if (arr1[i] < arr2[j]) {
10             merged[k++] = arr1[i++];
11         } else {
12             merged[k++] = arr2[j++];
13         }
14     }
15
16     while (i < n1) {
17         merged[k++] = arr1[i++];
18     }
19
20     while (j < n2) {
21         merged[k++] = arr2[j++];
22     }
23 }
24
25 int main() {
26     int arr1[] = {1, 3, 5, 7};
27     int n1 = sizeof(arr1) / sizeof(arr1[0]);
28
29     int arr2[] = {2, 4, 6, 8};
30     int n2 = sizeof(arr2) / sizeof(arr2[0]);
31
32     int merged[n1 + n2];
33
34     mergeArrays(arr1, n1, arr2, n2, merged);
35
36     printf("Merged array: ");
37     for (int i = 0; i < n1 + n2; i++) {
38         printf("%d ", merged[i]);
39     }
40
41     printf("\n");
42     return 0;
43 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q5.c -o q5
Merged array: 1 2 3 4 5 6 7 8
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```

C q1.c M C q2.c M C q3.c M C q4.c M C q5.c M C q6.c
assignment-3-sshahi7417-blip > src > C q6.c > main()
1 // Write a C program to find the length of a string without using the built-in string functions.
2 #include <stdio.h>
3
4 int main() {
5     char str[100];
6     int length = 0;
7
8     printf("Enter a string: ");
9     fgets(str, sizeof(str), stdin);
10
11     while (str[length] != '\0') {
12         length++;
13     }
14
15     if (length > 0 && str[length - 1] == '\n') {
16         length--;
17     }
18
19     printf("The length of the string is: %d\n", length);
20
21     return 0;
22 }
```

Output

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Merged array: 1 2 3 4 5 6 7 8
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q6.c -o q6
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q6
Enter a string: 4
The length of the string is: 1
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```

2
3  #include <stdio.h>
4
5  int main() {
6      char str[100];
7      int start = 0, end = 0;
8      char temp;
9
10     printf("Enter a string: ");
11     fgets(str, sizeof(str), stdin);
12
13     while (str[end] != '\0' && str[end] != '\n') {
14         end++;
15     }
16
17     end--;
18
19     while (start < end) {
20         temp = str[start];
21         str[start] = str[end];
22         str[end] = temp;
23
24         start++;
25         end--;
26     }
27
28     printf("Reversed string: %s\n", str);
29
30     return 0;
31 }
32

```

Output

```

the length of the string is: 1
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q7.c -o q7
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q7
Enter a string: 7
Reversed string: 7

PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src>

```

```
View Go Run Terminal Help c assingement 3
... C q1.c M C q2.c M C q3.c M C q4.c M C q5.c M C q6.c M C q7.c M C q8.c M X
assignment-3-sshahi7417-blip > src > C q8.c > main()
1 // Write a C program to check if a given string is a palindrome.
2 #include <stdio.h>
3 #include <stdbool.h>
4
5 int main() {
6     char str[100];
7     int length = 0;
8     bool isPalindrome = true;
9
10    printf("Enter a string: ");
11    fgets(str, sizeof(str), stdin);
12
13    while (str[length] != '\0' && str[length] != '\n') {
14        length++;
15    }
16
17    for (int i = 0; i < length / 2; i++) {
18        if (str[i] != str[length - 1 - i]) {
19            isPalindrome = false;
20            break;
21        }
22    }
23
24    if (isPalindrome) {
25        printf("The string is a palindrome.\n");
26    } else {
27        printf("The string is not a palindrome.\n");
28    }
29
30    return 0;
31 }
```

Output

```
U PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src>
M PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q8.c -o q8
U PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q8
M Enter a string: 234556
U The string is not a palindrome.
U PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> 
```



```
assignment-3-sshahi7417-blip > src > C q9.c > main()
1 // Implement a C program to count the occurrence of a specific character in a string.
2 #include <stdio.h>
3
4 int main() {
5     char str[100];
6     char target;
7     int count = 0;
8
9     printf("Enter a string: ");
10    fgets(str, sizeof(str), stdin);
11
12    printf("Enter a character to search for: ");
13    scanf("%c", &target);
14
15    for (int i = 0; str[i] != '\0' && str[i] != '\n'; i++) {
16        if (str[i] == target) {
17            count++;
18        }
19    }
20
21    printf("The character '%c' appears %d times.\n", target, count);
22
23    return 0;
24 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q9.c -o q9
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q9
Enter a string: 24
Enter a character to search for: r
The character 'r' appears 0 times.
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```
assignment-3-sshahi7417-blip > src > C q10.c > main()
1 // Write a C program to concatenate two strings without using the built-in string functions.
2 #include <stdio.h>
3
4 int main() {
5     char str1[100] = "Hello ";
6     char str2[] = "World!";
7     int i = 0, j = 0;
8
9     // Move i to the end of the first string
10    while (str1[i] != '\0') {
11        i++;
12    }
13
14    // Copy characters from str2 into str1 starting at index i
15    while (str2[j] != '\0') {
16        str1[i] = str2[j];
17        i++;
18        j++;
19    }
20
21    // Add the null terminator to the end of the combined string
22    str1[i] = '\0';
23
24    printf("Concatenated string: %s\n", str1);
25
26    return 0;
27 }
```

Output

```

PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q10.c -o q10
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q10
Concatenated string: Hello World!
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src>

```

```

j2.c  M  C q3.c  M  C q4.c  M  C q5.c  M  C q6.c  M  C q7.c  M  C q8.c  M  C q9.c  M  C q10.c  M  C q
assignment-3-sshahi7417-blip > src > C q11.c > main()
1 // Write a function named calculateAverage that takes an array of integers as input and returns the average of the numbers.
2
3 #include <stdio.h>
4
5 double calculateAverage(int arr[], int size) {
6     if (size <= 0) {
7         return 0.0;
8     }
9
10    int sum = 0;
11    for (int i = 0; i < size; i++) {
12        sum += arr[i];
13    }
14
15    return (double)sum / size;
16 }
17
18 int main() {
19     int numbers[] = {10, 20, 30, 40, 50};
20     int n = sizeof(numbers) / sizeof(numbers[0]);
21
22     double avg = calculateAverage(numbers, n);
23
24     printf("The average is: %.2f\n", avg);
25
26     return 0;
27 }

```

Output

```

Concatenated string: Hello World!
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q11.c -o q11
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q11
The average is: 30.00
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src>

```

```
assignment-3-sshahi7417-blip > src > C q12.c > main()
1 // Write a function named isPalindrome that takes a string as input and returns 1 if it is a palindrome (r
2
3 #include <stdio.h>
4
5 int isPalindrome(char str[]) {
6     int length = 0;
7     while (str[length] != '\0' && str[length] != '\n') {
8         length++;
9     }
10
11     for (int i = 0; i < length / 2; i++) {
12         if (str[i] != str[length - 1 - i]) {
13             return 0;
14         }
15     }
16     return 1;
17 }
18
19 int main() {
20     char input[100];
21     printf("Enter a string: ");
22     fgets(input, sizeof(input), stdin);
23
24     if (isPalindrome(input)) {
25         printf("1 (It is a palindrome)\n");
26     } else {
27         printf("0 (It is not a palindrome)\n");
28     }
29
30     return 0;
31 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q12.c -o q12
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q12
Enter a string: 71035
0 (It is not a palindrome)
```

```
Go Run Terminal Help ← → c assingement 3
4.c M C q5.c M C q6.c M C q7.c M C q8.c M C q9.c M
assignment-3-sshahi7417-blip > src > C q13.c > main()
1 // Write a function named findFactorial that takes an integer as input and ret
2
3 #include <stdio.h>
4
5 long long findFactorial(int n) {
6     if (n < 0) {
7         return -1;
8     }
9
10    long long factorial = 1;
11    for (int i = 1; i <= n; i++) {
12        factorial *= i;
13    }
14
15    return factorial;
16 }
17
18 int main() {
19     int num;
20     printf("Enter a positive integer: ");
21     scanf("%d", &num);
22
23     long long result = findFactorial(num);
24
25     if (result == -1) {
26         printf("Factorial of negative numbers is undefined.\n");
27     } else {
28         printf("Factorial of %d is %lld\n", num, result);
29     }
30
31     return 0;
32 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q13.c -o q13
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q13
Enter a positive integer: 12
Factorial of 12 is 479001600
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```
Go Run Terminal Help  ← → c assingement 3
C q5.c M C q6.c M C q7.c M C q8.c M C q9.c M C q10.c M C q
assignment-3-sshahi7417-blip > src > C q14.c > main()
1 // Write a function named convertTemperature that takes a temperature value in Celsius and
2
3
4 #include <stdio.h>
5
6 float convertTemperature(float celsius) {
7     return (celsius * 9 / 5) + 32;
8 }
9
10 int main() {
11     float celsius;
12
13     printf("Enter temperature in Celsius: ");
14     scanf("%f", &celsius);
15
16     float fahrenheit = convertTemperature(celsius);
17
18     printf("%.2f Celsius is equal to %.2f Fahrenheit\n", celsius, fahrenheit);
19
20     return 0;
21 }
```

Output

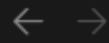
```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip> gcc q14.c -o q14
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip> ./q14
Enter temperature in Celsius: 17
17.00 Celsius is equal to 62.60 Fahrenheit
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip> |
```

```
6.c M C q7.c M C q8.c M C q9.c M C q10.c M C q11.c M C q12.c M
assignment-3-sshahi7417-blip > src > C q15.c > main()
1 // Write a function named countOccurrences that takes a string and a character as input and returns the number of occurrences of that character in the string.
2
3 #include <stdio.h>
4
5 int countOccurrences(char str[], char target) {
6     int count = 0;
7
8     for (int i = 0; str[i] != '\0'; i++) {
9         if (str[i] == target) {
10             count++;
11         }
12     }
13
14     return count;
15 }
16
17 int main() {
18     char text[] = "programming in c is fun!";
19     char search = 'n';
20
21     int result = countOccurrences(text, search);
22
23     printf("The character '%c' appears %d times in the string.\n", search, result);
24
25     return 0;
26 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q15.c -o q15
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q15
The character 'n' appears 3 times in the string.
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

Go Run Terminal Help



c assingement 3

C q7.c M

C q8.c M

C q9.c M

C q10.c M

C q11.c M

assignment-3-sshahi7417-blip > src > C q16.c > main()

```
3  #include <stdio.h>
4
5  void reverseArray(int arr[], int size) {
6      int start = 0;
7      int end = size - 1;
8      int temp;
9
10     while (start < end) {
11         temp = arr[start];
12         arr[start] = arr[end];
13         arr[end] = temp;
14
15         start++;
16         end--;
17     }
18 }
19
20 int main() {
21     int numbers[] = {1, 2, 3, 4, 5};
22     int n = sizeof(numbers) / sizeof(numbers[0]);
23
24     reverseArray(numbers, n);
25
26     printf("Reversed array: ");
27     for (int i = 0; i < n; i++) {
28         printf("%d ", numbers[i]);
29     }
30
31     printf("\n");
32     return 0;
33 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q16
Reversed array: 5 4 3 2 1
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> █
```

```
assignment-3-sshahi7417-blip / src / q17.c / main()
1 // Write a function named calculatePower that takes two integers, base and exponent, a
2
3 #include <stdio.h>
4
5 long long calculatePower(int base, int exponent) {
6     long long result = 1;
7     if (exponent == 0) {
8         return 1;
9     }
10
11     for (int i = 0; i < exponent; i++) {
12         result *= base;
13     }
14
15     return result;
16 }
17
18 int main() {
19     int b, e;
20
21     printf("Enter base: ");
22     scanf("%d", &b);
23     printf("Enter exponent: ");
24     scanf("%d", &e);
25
26     long long power = calculatePower(b, e);
27
28     printf("%d raised to the power of %d is %lld\n", b, e, power);
29
30     return 0;
31 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q17
Enter base: 9
Enter exponent: 4
9 raised to the power of 4 is 6561
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> █
```



```

assignment-3-sshahi7417-blip > src > C q18.c > main()
1 // Write a function named findPrimeNumbers that takes an integer n as i
2
3 #include <stdio.h>
4
5 void findPrimeNumbers(int n) {
6     for (int num = 2; num <= n; num++) {
7         int isPrime = 1;
8
9         for (int i = 2; i * i <= num; i++) {
10            if (num % i == 0) {
11                isPrime = 0;
12                break;
13            }
14        }
15
16        if (isPrime) {
17            printf("%d ", num);
18        }
19    }
20    printf("\n");
21 }
22
23 int main() {
24     int limit;
25     printf("Enter the limit: ");
26     scanf("%d", &limit);
27
28     printf("Prime numbers up to %d are: ", limit);
29     findPrimeNumbers(limit);
30
31     return 0;
32 }

```

Output

```

PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q18.c -o q18
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q18
Enter the limit: 100
Prime numbers up to 100 are: 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src>

```

```
q10.c M C q11.c M C q12.c M C q13.c M C q14.c M C q15
assignment-3-sshahi7417-blip > src > C q19.c > main()
1 // Write a function named calculateFactorialSeries that takes an integer n
2 #include <stdio.h>
3
4 void calculateFactorialSeries(int n) {
5     if (n < 1) {
6         printf("Please enter a positive integer.\n");
7         return;
8     }
9
10    long long fact = 1;
11    for (int i = 1; i <= n; i++) {
12        fact *= i;
13        printf("%d! = %lld\n", i, fact);
14    }
15 }
16
17 int main() {
18     int n;
19     printf("Enter the value of n: ");
20     scanf("%d", &n);
21
22     printf("Factorial series up to %d:\n", n);
23     calculateFactorialSeries(n);
24
25     return 0;
26 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q19.c -o q19
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q19
Enter the value of n: 6
Factorial series up to 6:
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
6! = 720
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```
Go Run Terminal Help c assingement 3
C q11.c M C q12.c M C q13.c M C q14.c M C q15.c M C q16.c M C q17.c M
assignment-3-sshahi7417-blip > src > C q20.c > main()
1 // Write a function named calculateGCD that takes two integers as input and returns their
2 #include <stdio.h>
3
4 int calculateGCD(int a, int b) {
5     int temp;
6     while (b != 0) {
7         temp = b;
8         b = a % b;
9         a = temp;
10    }
11    return a;
12 }
13
14 int main() {
15     int num1 = 48, num2 = 18;
16     printf("The GCD of %d and %d is: %d\n", num1, num2, calculateGCD(num1, num2));
17     return 0;
18 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip> gcc q20.c -o q20
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip> ./q20
The GCD of 48 and 18 is: 6
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip>
```

github-classroom[bot] (1 month ago)

```
Go Run Terminal Help  ← → c assingement 3
C q12.c M  C q13.c M  C q14.c M  C q15.c M  C q16.c M  C q17.c M
assignment-3-sshahi7417-blip > src > C q21.c > calculateFactorial(int)
1 // Write a recursive function named calculateFactorial that takes an integer n as in
2
3 #include <stdio.h>
4
5 long long calculateFactorial(int n) {
6     if (n <= 1) {
7         return 1;
8     }
9
10    return n * calculateFactorial(n - 1);
11 }
12
13 int main() {
14     int number = 5;
15
16     printf("Factorial of %d is %lld\n", number, calculateFactorial(number));
17
18     return 0;
19 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q21.c -o q21
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q21
Factorial of 5 is 120
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> 
```

 github-classroom[bot] (1 month ago) L

```
assignment-3-sshahi7417-blip > src > C q22.c > main()
1 // Write a recursive function named calculateFibonacci that takes an integer n as input and returns the nth Fibonacci number.
2
3 #include <stdio.h>
4
5 long long memo[100];
6
7 long long calculateFibonacci(int n) {
8     if (n <= 0) return 0;
9     if (n == 1) return 1;
10
11     if (memo[n] != 0) {
12         return memo[n];
13     }
14
15     memo[n] = calculateFibonacci(n - 1) + calculateFibonacci(n - 2);
16     return memo[n];
17 }
18
19 int main() {
20     int n = 50;
21
22     printf("The %dth Fibonacci number is: %lld\n", n, calculateFibonacci(n));
23
24     return 0;
25 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q22.c -o q22
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q22
The 50th Fibonacci number is: 12586269025
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> █
```

```
assignment-3-sshahi7417-blip > src > C q23.c > main()
1 // Write a recursive function named calculateGCD that takes two integers a and b as input a
2
3 #include <stdio.h>
4
5 int calculateGCD(int a, int b) {
6     if (b == 0) {
7         return a;
8     }
9
10    return calculateGCD(b, a % b);
11
12 }
13
14 int main() {
15     int num1 = 48, num2 = 18;
16
17     printf("The GCD of %d and %d is: %d\n", num1, num2, calculateGCD(num1, num2));
18
19     return 0;
20 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q23.c -o q23
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q23
The GCD of 48 and 18 is: 6
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
github-classroom[bot] (1 mont
```

```
view Go Run Terminal Help c assingement 3
q15.c M C q16.c M C q17.c M C q18.c M C q19.c M C q20.c M C q21.c M C q22.c M
assignment-3-sshahi7417-blip > src > C q24.c > main()
1 // Write a recursive function named calculatePower that takes two integers base and exponent as input and retu
2
3 #include <stdio.h>
4
5 long long calculatePower(int base, int exponent) {
6     if (exponent == 0) {
7         return 1;
8     }
9     return base * calculatePower(base, exponent - 1);
10 }
11
12 int main() {
13     int b = 2, e = 10;
14     printf("%d raised to the power of %d is %lld\n", b, e, calculatePower(b, e));
15     return 0;
16 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q24.c -o q24
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q24
2 raised to the power of 10 is 1024
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```
assignment-3-sshahi7417-blip > src > C q25.c > main()
1 // Write a recursive function named calculateSumOfDigits that takes an integer
2 #include <stdio.h>
3
4 int calculateSumOfDigits(int n) {
5     if (n == 0) {
6         return 0;
7     }
8
9     return (n % 10) + calculateSumOfDigits(n / 10);
10 }
11
12 int main() {
13     int num = 1234;
14     printf("Sum of digits of %d is %d\n", num, calculateSumOfDigits(num));
15     return 0;
16 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q25.c -o q25
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q25
Sum of digits of 1234 is 10
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

Not Comm

```
o Run Terminal Help ← → c assingement 3
.c M C q18.c M C q19.c M C q20.c M C q21.c M
assignment-3-sshahi7417-blip > src > C q26.c > main()
1 // Write a recursive function named reverseString that takes a
2 #include <stdio.h>
3 #include <string.h>
4
5 void reverseString(char *str) {
6
7     if (*str == '\0') {
8         return;
9     }
10
11     reverseString(str + 1);
12     printf("%c", *str);
13 }
14
15 int main() {
16     char str[] = "Hello World";
17     printf("Original: %s\n", str);
18     printf("Reversed: ");
19     reverseString(str);
20     printf("\n");
21     return 0;
22 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q26.c -o q26
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q26
Original: Hello World
Reversed: dlrow olleH
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```


8.c	M	C q19.c	M	C q20.c	M	C q21.c	M	C q22.c	M
-----	---	---------	---	---------	---	---------	---	---------	---

```
assignment-3-sshahi7417-blip > src > C q27.c > main()
1 // Write a recursive function named printTriangle that takes
2 #include <stdio.h>
3
4 void printRow(int m) {
5     if (m == 0) {
6         return;
7     }
8     printf("* ");
9     printRow(m - 1);
10 }
11
12 void printTriangle(int n) {
13     if (n <= 0) {
14         return;
15     }
16
17     printTriangle(n - 1);
18     printRow(n);
19     printf("\n");
20 }
21
22 int main() {
23     int rows = 5;
24     printTriangle(rows);
25     return 0;
26 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q27.c -o q27
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q27
*
* *
* * *
* * * *
* * * * *
```

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> |
```

```

c M X C q20.c M C q21.c M C q22.c M C q23.c M C q24
assignment-3-sshahi7417-blip > src > C q28.c > main()
1 // Write a recursive function named calculateBinary that takes an integ
2 #include <stdio.h>
3 #include <string.h>
4
5 void convertToBinary(int n, char *buffer, int *index) {
6
7     if (n > 1) {
8         convertToBinary(n / 2, buffer, index);
9     }
10
11     buffer[(*index)++] = (n % 2) + '0';
12     buffer[*index] = '\0';
13 }
14
15 int main() {
16     int n = 13;
17     char binaryString[33];
18     int index = 0;
19
20     if (n == 0) {
21         strcpy(binaryString, "0");
22     } else {
23         convertToBinary(n, binaryString, &index);
24     }
25
26     printf("Decimal: %d\nBinary: %s\n", n, binaryString);
27     return 0;
28 }

```

Output

```

PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q28.c -o q28
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q28
Decimal: 13
Binary: 1101
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src>

```

M	C q21.c M	C q22.c M	C q23.c M	C q24.c M	C q25.c M
---	-----------	-----------	-----------	-----------	-----------

```
assignment-3-sshahi7417-blip > src > C q29.c > main()
1 // Write a recursive function named isPalindrome that takes a string as input
2 #include <stdio.h>
3 #include <string.h>
4
5 int checkPalindrome(char *str, int left, int right) {
6     if (left >= right) {
7         return 1;
8     }
9
10    if (str[left] != str[right]) {
11        return 0;
12    }
13    return checkPalindrome(str, left + 1, right - 1);
14 }
15
16 int isPalindrome(char *str) {
17     int len = strlen(str);
18     if (len == 0) return 1;
19     return checkPalindrome(str, 0, len - 1);
20 }
21
22 int main() {
23     char str[] = "radar";
24     if (isPalindrome(str)) {
25         printf("%s' is a palindrome.\n", str);
26     } else {
27         printf("%s' is not a palindrome.\n", str);
28     }
29     return 0;
30 }
```

Output

```
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q29.c -o q29
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q29
'radar' is a palindrome.
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> █
```

```

assignment-3-sshahi7417-blip > src > C q30.c > main()
1 // Write a recursive function named countOccurrences that takes a string
2 #include <stdio.h>
3
4 int countOccurrences(char *str, char target) {
5
6     if (*str == '\0') {
7         return 0;
8     }
9
10    if (*str == target) {
11        return 1 + countOccurrences(str + 1, target);
12    } else {
13        return countOccurrences(str + 1, target);
14    }
15 }
16
17 int main() {
18     char myString[] = "banana";
19     char searchChar = 'a';
20
21     int result = countOccurrences(myString, searchChar);
22
23     printf("The character '%c' appears %d times in \"%s\".\n",
24         searchChar, result, myString);
25
26     return 0;
27 }

```

Output

```

PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> gcc q30.c -o q30
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src> ./q30
The character 'a' appears 3 times in "banana".
PS C:\Users\Smriti Shahi\Downloads\c assingement 3\assignment-3-sshahi7417-blip\src>

```